CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 1: Advanced Relational Algebra & Complex SQL Queries

Course: Advanced Database Engineering [23CS3503]
Category: Assignments | Assignment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Deep dive into formal relational algebra and advanced analytical SQL queries.

2. Assignment Problems & Instructions:

1. Translate complex natural language queries into extended Relational Algebra expressions.
2. Write analytical window functions using ROW_NUMBER , RANK , and DENSE_RANK .
3. Construct hierarchical recursive Common Table Expressions CTEs for employee trees.
4. Optimize multi-table joins using appropriate Cartesian product filtering rules.

Submission Guidelines & Verification:

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